

2008 ACCESSORIES & BODY, CAB

Horn - Edge & MKX

SPECIFICATIONS

TORQUE SPECIFICATIONS

| Description | Nm | lb-in |
|-------------|----|-------|
| Horn bolt   | 10 | 89    |

DESCRIPTION AND OPERATION

HORN

**NOTE:**       The smart junction box (SJB) is also known as the generic electronic module (GEM).

The horn system consists of the following components:

- Horn switches (part of the steering wheel)
- Clockspring
- Horn relay (part of the SJB)
- Horn

For removal and installation of the steering wheel, refer to STEERING COLUMN article.

For removal and installation of the SJB, refer to MULTIFUNCTION ELECTRONIC MODULES article.

Horn Location

The horn is mounted on the side of the RH wheel well under the front fender splash shield.

DIAGNOSTIC TESTS

HORN

Special Tools

| Illustration                                                                                            | Tool Name              | Tool Number             |
|---------------------------------------------------------------------------------------------------------|------------------------|-------------------------|
|  <div>ST1137-A</div> | 73III Automotive Meter | 105-R0057 or equivalent |
|                                                                                                         |                        |                         |

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|                                                                                               |                                                                                                                                       |                         |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <br>ST2834-A | Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool |                         |
| <br>ST2574-A | Flex Probe Kit                                                                                                                        | 105-R025C or equivalent |

### Principles of Operation

**NOTE:** The smart junction box (SJB) is also known as the generic electronic module (GEM).

The SJB supplies the control and switched voltage to the horn relay (part of the SJB). When the horn switches (part of the steering wheel) are pressed, ground is supplied through the clockspring to the horn relay. The horn relay is then energized, directing voltage to the horn and enabling the horn to sound.

The horn switch is comprised of 2 sets of contacts separated by springs. The lower set is connected to ground and the upper set is connected to the horn signal circuit. When the driver air bag module is pressed, it pushes down on the upper set of contacts, collapsing the springs and allowing the contacts to touch. When the contacts touch, it completes the circuit.

When the vehicle security system is armed, the SJB provides a momentary ground to the horn relay control side to indicate the system is armed, and an on and off ground when an intrusion is detected or the panic alarm is activated.

### Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

### VISUAL INSPECTION CHART

| Mechanical                                                                                                                                          | Electrical                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Horn</li><li>• Steering wheel switch harness</li><li>• Horn switches (part of the steering wheel)</li></ul> | <ul style="list-style-type: none"><li>• Smart junction box (SJB) fuse 24 (20A)</li><li>• Wiring, terminals or connectors</li><li>• Clockspring</li><li>• SJB</li></ul> |

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3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

**NOTE:**        **Make sure to use the latest scan tool release software.**

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

**NOTE:**        **The vehicle communication module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM.**

5. If the scan tool does not communicate with the VCM:
  - Check the VCM connection to the vehicle.
  - Check the scan tool connection to the VCM.
  - Refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the vehicle.
6. If the scan tool does not communicate with the vehicle:
  - Verify the ignition key is in the ON position.
  - Verify the scan tool operation with a known good vehicle.
  - Refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
  - If the scan tool responds with no communication with one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
  - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the SJB.
9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
10. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

#### Symptom Chart

#### Symptom Chart

| Condition                                                                 | Possible Sources                                                                                                                                                                                                                   | Action                                                                           |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• The horn is inoperative</li></ul> | <ul style="list-style-type: none"><li>• Fuse</li><li>• Wiring, terminals or connectors</li><li>• Horn</li><li>• Clockspring</li><li>• Steering wheel switch harness</li><li>• Horn switches (part of the steering wheel)</li></ul> | <ul style="list-style-type: none"><li>• Go to <b>Pinpoint Test A</b> .</li></ul> |

|                                                                         |                                                                                                                                                                                                                    |                                                                                  |
|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                                                                         | <ul style="list-style-type: none"><li>• Smart junction box (SJB)</li></ul>                                                                                                                                         |                                                                                  |
| <ul style="list-style-type: none"><li>• The horn is always on</li></ul> | <ul style="list-style-type: none"><li>• Wiring, terminals or connectors</li><li>• Clockspring</li><li>• Steering wheel switch harness</li><li>• Horn switches (part of the steering wheel)</li><li>• SJB</li></ul> | <ul style="list-style-type: none"><li>• Go to <b>Pinpoint Test B</b> .</li></ul> |

**Pinpoint Tests**

**Pinpoint Test A: The Horn Is Inoperative**

Refer to appropriate SYSTEM WIRING DIAGRAMS article , Horn/Cigar Lighter for schematic and connector information.

**Normal Operation**

The smart junction box (SJB) supplies the switched and control voltage to the horn relay (part of the SJB). When the horn switches (part of the steering wheel) are pressed, the switches are grounded to the steering wheel through circuit GD165 (BK), and the clockspring to the instrument panel cross-car beam through circuit GD116 (BK/VT). Ground is then supplied to the horn relay through circuit CRH02 (BU/WH), energizing the horn relay. The horn relay then sends voltage to the horn through circuit CRH04 (VT/GN), enabling the horn to sound. Ground for the horn is supplied through circuit GD123 (BK/GY).

**This pinpoint test is intended to diagnose the following:**

- Fuse
- Wiring, terminals or connectors
- Horn
- Clockspring
- Steering wheel switch harness
- Horn switches (part of the steering wheel)
- SJB

**PINPOINT TEST A: THE HORN IS INOPERATIVE**

**CAUTION:** Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

**A1 CHECK THE SJB OUTPUT**

- Key in ON position.

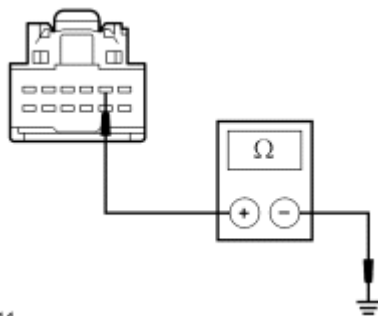
- Enter the following diagnostic mode on the diagnostic tool: SJB DataLogger
- Select the SJB PID (HORN). Active command the horn on and then off.
- **Does the horn sound when commanded on?**

**YES** : Go to A2.

**NO** : Go to A7.

## A2 CHECK CIRCUIT GD165 (BK) FOR AN OPEN

- Key in OFF position.
- Remove the driver air bag module. Refer to **SUPPLEMENTAL RESTRAINT SYSTEM** article.
- Disconnect: Clockspring C218b



N0026511

**Fig. 1: Checking Circuit GD165 (BK) For Open**  
Courtesy of FORD MOTOR CO.

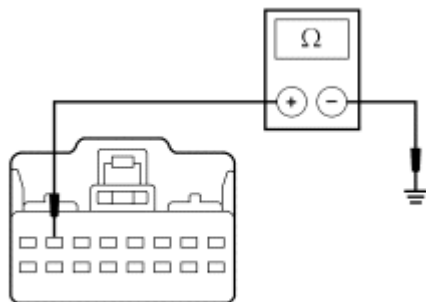
- Measure the resistance between the clockspring C218b-2, circuit GD165 (BK), harness side and ground.
- **Is the resistance less than 5 ohms?**

**YES** : Go to A3.

**NO** : REPAIR the circuit. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

## A3 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN

- Disconnect: Clockspring C218a



N0026773

**Fig. 2: Checking Circuit GD116 (BK/VT) For Open**  
Courtesy of FORD MOTOR CO.

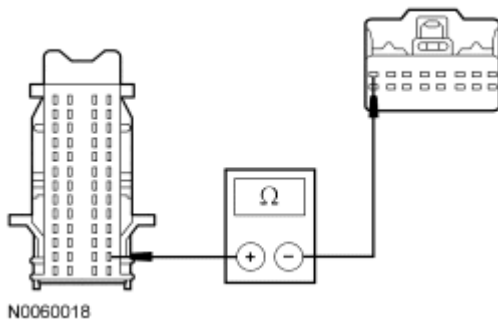
- Measure the resistance between the clockspring C218a-7, circuit GD116 (BK/VT), harness side and ground.
- **Is the resistance less than 5 ohms?**

**YES :** Go to A4.

**NO :** REPAIR the circuit. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

#### **A4 CHECK CIRCUIT CRH02 (BU/WH) FOR AN OPEN**

- Disconnect: SJB C2280b



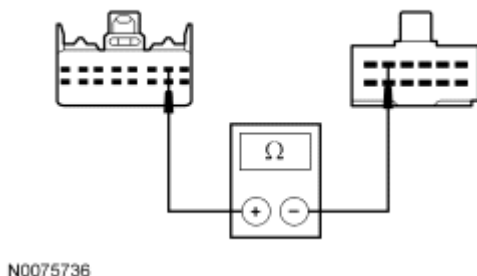
**Fig. 3: Checking Circuit CRH02 (BU/WH) For Open**  
Courtesy of FORD MOTOR CO.

- Measure the resistance between the clockspring C218a-8, circuit CRH02 (BU/WH), harness side and the SJB C2280b-41, circuit CRH02 (BU/WH), harness side.
- **Is the resistance less than 5 ohms?**

**YES :** Go to A5.

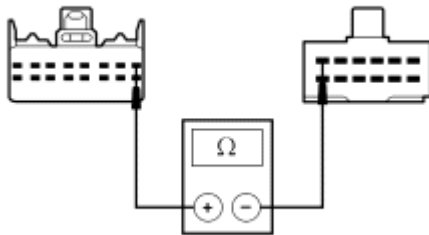
**NO :** REPAIR the circuit. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

#### **A5 CHECK THE CLOCKSPRING FOR AN OPEN**



**Fig. 4: Checking Clockspring For Open**  
Courtesy of FORD MOTOR CO.

- Measure the resistance between the clockspring C218a pin 7, component side and the clockspring C218b pin 2, component side.



N0075737

**Fig. 5: Checking Clockspring For Open**  
Courtesy of FORD MOTOR CO.

- Measure the resistance between the clockspring C218a pin 8, component side and the clockspring C218b pin 1, component side.

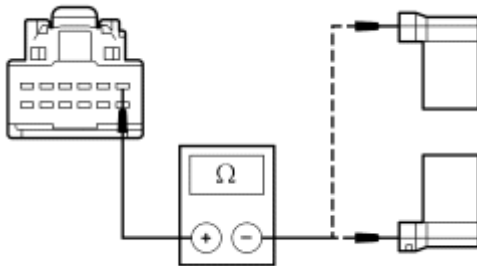
- **Are the resistances less than 5 ohms?**

**YES :** Go to A6.

**NO :** INSTALL a new clockspring. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

#### **A6 CHECK THE STEERING WHEEL SWITCH HARNESS FOR AN OPEN**

- Disconnect: Horn Switch C2996 and C2997



N0026922

**Fig. 6: Checking Steering Wheel Switch Harness For Open**  
Courtesy of FORD MOTOR CO.

- Measure the resistance between the clockspring C218b-1, circuit CRH02 (BU/WH), harness side and the horn switch C2996-1, circuit CRH02 (BU/WH), harness side; and between the clockspring C218b-1, circuit CRH02 (BU/WH), harness side and the horn switch C2997-1, circuit CRH02 (BU/WH), harness side.

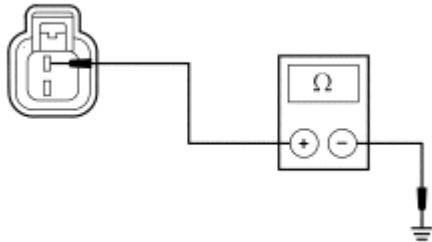
- **Are the resistances less than 5 ohms?**

**YES :** INSTALL a new steering wheel. REFER to **STEERING COLUMN** article. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

**NO :** REPAIR the circuit. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

**A7 CHECK CIRCUIT GD123 (BK/GY) FOR AN OPEN**

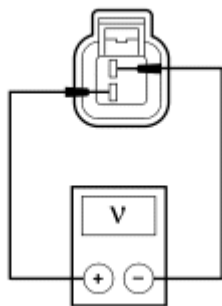
- Key in OFF position.
- Disconnect: Horn C131



A0032815

**Fig. 7: Checking Circuit GD123 (BK/GY) For Open**  
Courtesy of FORD MOTOR CO.

- Measure the resistance between the horn C131-1, circuit GD123 (BK/GY), harness side and ground.
- **Is the resistance less than 5 ohms?**  
**YES** : Go to A8.  
**NO** : REPAIR the circuit. TEST the system for normal operation.

**A8 CHECK CIRCUIT CRH04 (VT/GN) FOR VOLTAGE**

N0013171

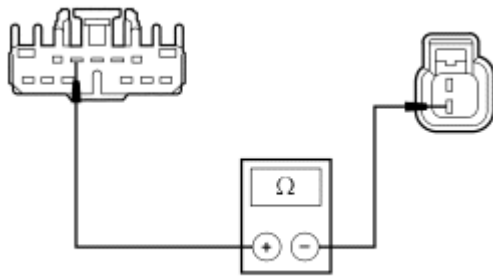
**Fig. 8: Checking Circuit CRH04 (VT/GN) For Voltage**  
Courtesy of FORD MOTOR CO.

- Measure the voltage between the horn C131-2, circuit CRH04 (VT/GN), harness side, and the horn C131-1, circuit GD123 (BK/GY), harness side, while pressing the horn switch.
- **Is the voltage greater than 10 volts?**  
**YES** : INSTALL a new horn. REFER to **Horn**. TEST the system for normal operation.  
**NO** : VERIFY the SJB fuse 24 (20A) is OK. If OK, go to A9.

**A9 CHECK CIRCUIT CRH04 (VT/GN) FOR AN OPEN**

- Disconnect: SJB C2280e





N0053603

**Fig. 9: Checking Circuit CRH04 (VT/GN) For Open**  
Courtesy of FORD MOTOR CO.

- Measure the resistance between the SJB C2280e-5 , circuit CRH04 (VT/GN), harness side, and the horn C131-2, circuit CRH04 (VT/GN), harness side.
- **Is the resistance less than 5 ohms?**  
**YES :** Go to A10.  
**NO :** REPAIR the circuit. TEST the system for normal operation.

#### **A10 CHECK FOR CORRECT SJB OPERATION**

- Disconnect all the SJB connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

**YES :** INSTALL a new SJB. REFER to **MULTIFUNCTION ELECTRONIC MODULES** article. TEST the system for normal operation.

**NO :** The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

#### **Pinpoint Test B: The Horn Is Always On**

Refer to appropriate SYSTEM WIRING DIAGRAMS article , Horn/Cigar Lighter for schematic and connector information.

#### **Normal Operation**

The smart junction box (SJB) supplies the switched and control voltage to the horn relay (part of the SJB). When the horn switches (part of the steering wheel) are pressed, the switches are grounded to the steering wheel, and the clockspring to the instrument panel cross-car beam. Ground is then supplied to the horn relay through circuit CRH02 (BU/WH), energizing the horn relay. The horn relay then sends voltage to the horn

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through circuit CRH04 (VT/GN), enabling the horn to sound.

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Clockspring
- Steering wheel switch harness
- Horn switches (part of the steering wheel)
- SJB

### **PINPOINT TEST B: THE HORN IS ALWAYS ON**

**CAUTION:** Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

#### **B1 CHECK CIRCUIT CRH04 (VT/GN) FOR A SHORT TO VOLTAGE**

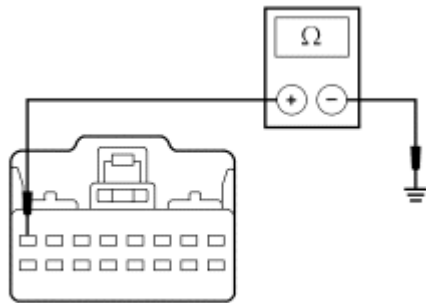
- Key in OFF position.
- Disconnect: SJB C2280e
- Key in ON position.
- **Does the horn continue to sound?**  
**YES :** REPAIR the circuit. TEST the system for normal operation.  
**NO :** Go to B2.

#### **B2 CHECK THE HORN SWITCH INPUT**

- Key in OFF position.
- Disconnect: SJB C2280b
- Connect: SJB C2280e
- **Does the horn sound continuously?**  
**YES :** Go to B6.  
**NO :** Go to B3.

#### **B3 CHECK CIRCUIT CRH02 (BU/WH) FOR A SHORT TO GROUND**

- Depower the supplemental restraint system (SRS). Refer to **SUPPLEMENTAL RESTRAINT SYSTEM** article.
- Disconnect: Clockspring C218a



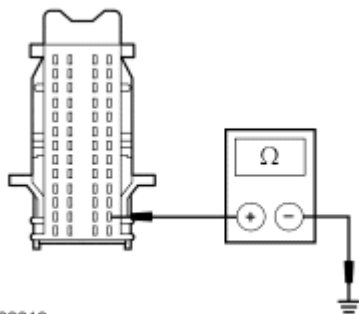
N0026775

**Fig. 10: Checking Circuit CRH02 (BU/WH) For Short To Ground**  
 Courtesy of FORD MOTOR CO.

- Measure the resistance between the clockspring C218a-8, circuit CRH02 (BU/WH), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**  
 YES : Go to B4.  
 NO : REPAIR the circuit. REPOWER the SRS. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

**B4 CHECK THE CLOCKSPRING FOR A SHORT TO GROUND**

- Remove the driver air bag module. Refer to **SUPPLEMENTAL RESTRAINT SYSTEM** article.
- Connect: Clockspring C218a
- Disconnect: Clockspring C218b



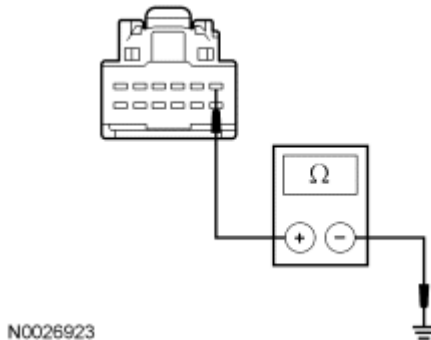
N0060019

**Fig. 11: Checking Clockspring For Short To Ground**  
 Courtesy of FORD MOTOR CO.

- Measure the resistance between the SJB C2280b-41, circuit CRH02 (BU/WH), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**  
 YES : Go to B5.  
 NO : INSTALL a new clockspring. INSTALL the driver air bag module. REFER to **SUPPLEMENTAL RESTRAINT SYSTEM** article. TEST the system for normal operation.

**B5 CHECK THE STEERING WHEEL SWITCH HARNESS FOR A SHORT TO GROUND**

- Disconnect: Horn Switch C2996 and C2997



**Fig. 12: Checking Steering Wheel Switch Harness For Short To Ground**  
Courtesy of FORD MOTOR CO.

- Measure the resistance between the clockspring 218b-1, circuit CRH02 (BU/WH), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**

**YES :** INSTALL a new steering wheel. REFER to STEERING COLUMN article. INSTALL the driver air bag module. REFER to SUPPLEMENTAL RESTRAINT SYSTEM article. TEST the system for normal operation.

**NO :** REPAIR the circuit. INSTALL the driver air bag module. REFER to SUPPLEMENTAL RESTRAINT SYSTEM article. TEST the system for normal operation.

#### **B6 CHECK FOR CORRECT SJB OPERATION**

- Disconnect all the SJB connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the SJB connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

**YES :** INSTALL a new SJB. REFER to MULTIFUNCTION ELECTRONIC MODULES article. TEST the system for normal operation.

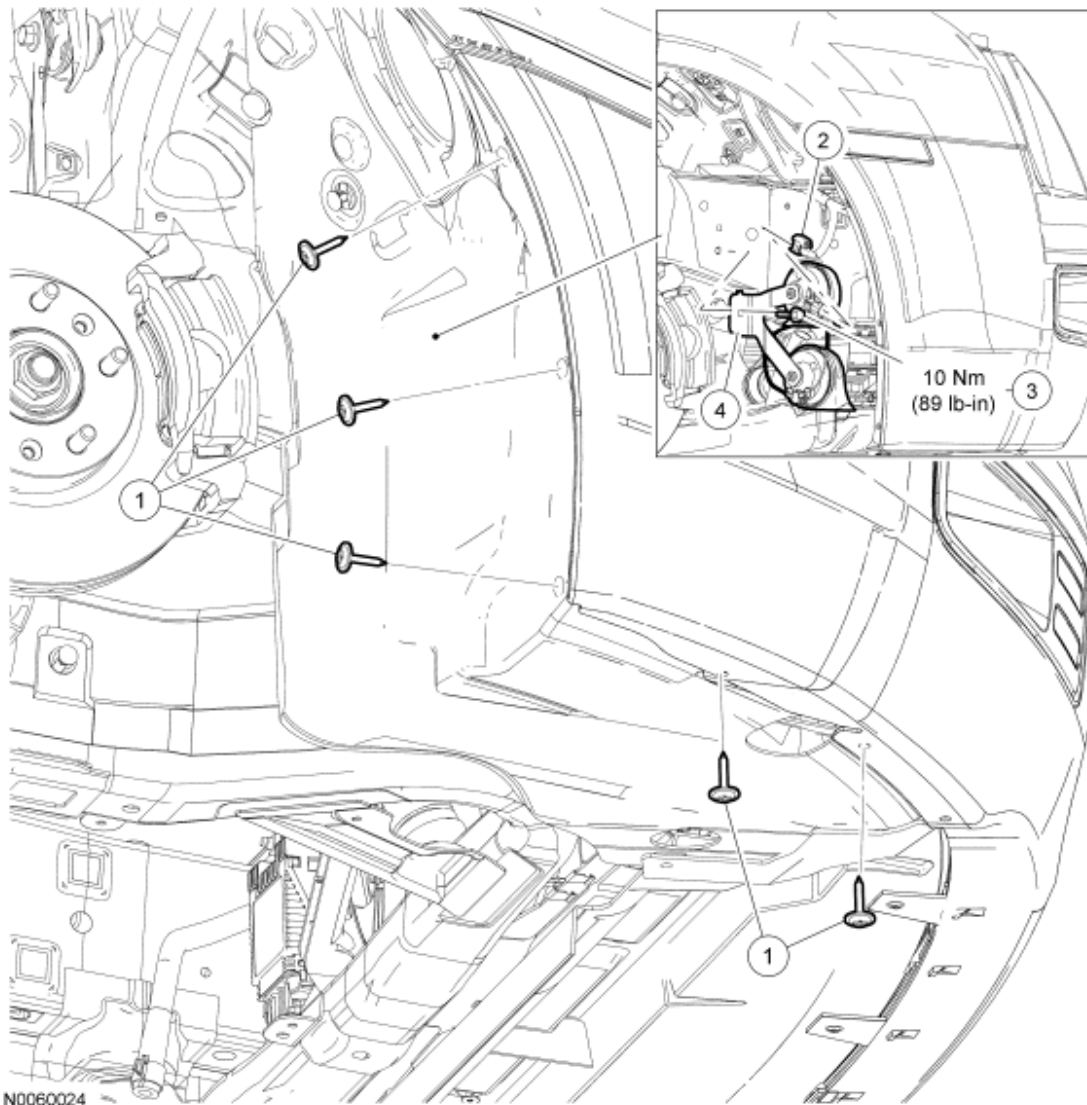
**NO :** The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

## **REMOVAL AND INSTALLATION**

### **HORN**

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**Fig. 13: Exploded View Of Horn Components (With Torque Specifications)**  
Courtesy of FORD MOTOR CO.

| Item | Part Number | Description                                         |
|------|-------------|-----------------------------------------------------|
| 1    | W706805     | Front fender splash shield screws (RH) (5 required) |
| 2    | -           | Horn electrical connector (part of 14290)           |
| 3    | W605784     | Horn bolt                                           |
| 4    | 13832       | Horn                                                |

#### REMOVAL AND INSTALLATION

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.

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2. Remove the 5 RH front fender splash shield screws.
  - Pull the RH front fender splash shield downward to access the horn.
3. Disconnect the electrical connector.
4. Remove the bolt and the horn.
  - To install, tighten to 10 Nm (89 lb-in).
5. To install, reverse the removal procedure.